Why a Rare Detour Can Dominate Global Optimization Time

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The Global Convergence Time of Stochastic Gradient Descent in Non-Convex Landscapes: Sharp Estimates via Large Deviations

Key: azizian2025global Tags: D / S / E; R / B Tier: Must read

Survey: Dynamics; Escape methods

Spreadsheet priority: None; independent recommendation: Must read

The paper in one minute

Question: How long does constant-step SGD take to enter a neighborhood of a global minimizer in a nonconvex landscape?

Model/Method: A large-deviation action defines transition costs between critical components; directed forests and prunings determine an energy exponent.

Finding: The paper reports exponential-scale expected hitting-time bounds of the form $\exp[(E \pm a)/\eta]$. Matching lower bounds require sufficiently strong target attraction.

Why we care: The mean time can be controlled by a rare detour into a deep trap. This differs from efficient saddle escape; an appendix proof step is explicitly flagged in these notes.

1. Reaching the global target is a different question

Finding a small gradient or escaping negative curvature does not ensure that a better basin has been reached. Let $Q = \arg\min f$ and define $\tau\delta = \inf\{n \geq 0 : \mathrm{dist}(x_n, Q) \leq \delta\}$. The paper asks about $E_x[\tau\delta]$, the mean first time constant-step SGD gets close to Q from x. It does not claim that every later iterate stays at a global minimum or that Q can be cheaply recognized in practice.

$$x_{n+1} = x_n - \eta G(x_n;\omega_n),\ G(x;\omega) = \nabla f(x) + Z(x;\omega).$$

Noise seeds are independent across iterations. The blanket objective assumptions are C^2 regularity, coercive loss and gradient norm at infinity, a globally Lipschitz gradient, and finitely many suitably connected critical components. Noise is unbiased, has positive-definite covariance, smooth controlled growth and a uniform sub-Gaussian bound. Finite-action transitions must connect the components. These are not automatic for arbitrary unregularized ReLU networks or every minibatch oracle.

2. A cost for unlikely trajectories

The Hamiltonian $H(x,p) = \log E \exp(-p \cdot G(x;\omega))$ encodes the full local noise distribution. Its convex conjugate $L(x,v)$ measures the cost of realizing velocity v. Integrating L along a path γ gives its action $S[\gamma]$. Downhill gradient flow has zero action; forcing a path against the drift generally costs positive action. Transition probabilities then have an exponential scale $\exp(-\text{cost}/\eta)$. This is a small-step asymptotic description, not an exact formula at any chosen learning rate.

3. Worked example: enumerate the graph, not just the shortest route

Pedagogical illustration, constructed for these notes.

Consider a pedagogical Markov model with non-target states A,B and absorbing target G. Assign direct transition costs A→G:2, A→B:3, B→A:5, B→G:8. A cost c means probability exp(-c/η); the remaining probability is a self-loop. All outgoing sums are below one for the numerical step sizes used below. This finite chain illustrates the graph mechanism, not a claimed neural-network construction.

A forest reaching G must select one outgoing edge from each non-target and contain no cycle. The three choices are {A→G,B→A}, {A→G,B→G}, and {A→B,B→G}, with costs 7,10,11. Thus the minimum forest cost is E(G) = 7.

A pruning has one edge and no path from A to G. The eligible edges are A→B, B→A, B→G, with costs 3,5,8. The minimum is 3, giving E(G|A) = 7 - 3 = 4. Merely looking at the direct cost A→G = 2 misses this exponent.

Write $u = e^{-2/\eta}$, $v = e^{-3/\eta}$, $w = e^{-5/\eta}$, $z = e^{-8/\eta}$. First-step conditioning gives $(u+v)T_a - vT_\beta = 1$ and $-wT_a + (w+z)T_\beta = 1$.

Solving yields $T_a = (w+z+v)/(uw+uz+vz)$. The leading numerator is $e^{-3/\eta}$; the leading denominator is $e^{-7/\eta}$. Hence T_a has scale $e^{4/\eta}$, confirming the forest calculation independently.

η	Exact T_a and logarithmic scale
0.5	$T_a \approx 3027.17$; $\eta \log T_a \approx 4.00769$.
0.25	$T_a \approx 8{,}889{,}035.88$; $\eta \log T_a \approx 4.00008$.

At A, the chance of choosing the rarer B detour before the direct target move has scale $e^{-1/\eta}$. Residence near B has scale $e^{5/\eta}$. Their product has scale $e^{4/\eta}$, which dominates the typical direct route. A mean can therefore be governed by rare, unusually long trajectories. Minimum-path closure of the direct costs leaves this example's exponent unchanged.

4. From critical components to the published exponent

Nodes represent connected critical components, not individual samples or parameter coordinates. B_{ij} is the least action of moving between components while avoiding the target beforehand. A transition forest directs every non-target node toward Q without cycles. A pruning deletes one outgoing edge so that the selected starting node p cannot reach Q. The energy difference is E(Q|p) = minimum forest cost - minimum pruning cost. For general x, the paper further adjusts for initial transition costs to possible traps.

$$E(Q|x) = \max_p [E(Q|p) - B(x,p)]_+.$$

5. State both sides with their qualifications

Theorem 1 reports, for sufficiently small target radius δ and step η and any logarithmic tolerance a > 0, the following bounds. The upper bound uses Assumptions 1-3. The matching lower bound additionally requires a sufficiently strong attracting neighborhood of Q; Appendix D.3 makes this requirement quantitative.

$$\exp[(E(Q|x)-a)/\eta] \leq E_x[\tau\delta] \leq \exp[(E(Q|x)+a)/\eta].$$

The tolerance a concerns the exponential rate, whereas δ concerns distance to the global target. Neither is the gradient tolerance ε of a second-order stationarity theorem. If E is positive, the mean time can be exponentially large in 1/η. If no spurious locally minimizing component exists, Proposition D.1 yields E = 0 and subexponential upper scaling. Subexponential does not mean polynomial or practical.

6. How the proof is organized

Appendix B states the path large-deviation principle. Appendix C constructs a chain of visits to critical neighborhoods and estimates its transition probabilities. Appendix D applies finite-state forest formulas to the induced chain, then translates its time scale back to actual SGD steps. Retaining enough attraction near Q is the extra step used for the lower bound. Our exact three-state calculation isolates the finite-chain algebra without assuming the delicate continuous-state reduction.

7. A Gaussian action makes the geometry intuitive

For an ideal additive Gaussian oracle with covariance $\sigma^2 I$, $H(x,p) = -p\cdot\nabla f(x) + \sigma^2\|p\|^2/2$ and $L(x,v) = \|v+\nabla f(x)\|^2/(2\sigma^2)$. Completing squares gives $L(x,v)-L(x,-v) = 2v\cdot\nabla f(x)/\sigma^2$. Along a reversed gradient-flow connection, uphill action is therefore $2\Delta f/\sigma^2$ while downhill action is zero. The general controlled-noise theorem uses its actual cumulant function; the Gaussian warmup does not remove the blanket regularity and growth requirements.

8. What the numerical evidence covers

Figure 2 uses a modified two-dimensional three-hump camel function. For each η the authors run 500 trajectories with Gaussian noise $\sigma = 50$ and target distance 10^{-2}, comparing log hitting times with inverse step size. Appendix F adds two-dimensional Styblinski-Tang and Himmelblau examples. These support the predicted scale on selected toy landscapes, not deep-network runtime or generalization. Polynomial test functions also exceed global Lipschitz smoothness at infinity, so local numerical agreement should be separated from blanket theorem coverage.

9. Source audit and limitations

The published lower-bound proof needs further justification at Lemma D.7. Its weighted sum M_n in (D.44) obeys $E[M_n|\text{past}] = (1-\mu\eta)M_{n-1}$, so it is generally not a martingale as asserted before the Doob-inequality step (D.46). The guide reports the lower-bound statement with its attraction condition, flags this unresolved proof step, and does not certify a repaired proof. Several other displayed sign/index slips should likewise not be copied into code. Knowing the graph and all transition costs can itself be hard; this is an analysis framework, not an efficient global optimizer.

10. Survey connection

Fang2019 addresses efficient approximate second-order stationarity. Gürbüzbalaban2021 addresses stationary heavy tails under streaming random curvature. Here the target is first entrance near an actual global optimum, with controlled-noise and geometric conditions. These different targets explain why an exponential hitting time does not contradict a polynomial local-stationarity bound.

Quantity	Do not confuse it with
τ_δ	Permanent convergence or a practical stopping test.
$E(Q\|x)$	Only the cheapest path to the target.
$E = 0$	A proved polynomial complexity bound.

Source: ICML/PMLR original PDF1-9; selected necessary appendices B-E and F-G. Central locators: Theorem1, Eq.13-23, D.1-D.3, PropositionD.1. Toy Markov example constructed and checked independently.

Papers Master - Our notes (copy): Large-deviation transition costs and forest/pruning energies describe small-step expected global-target hitting times. Matching lower bounds add a strong-attraction requirement; the guide flags an unresolved martingale step in that proof. This is not an efficient global optimizer, and subexponential scaling does not imply polynomial time.

Reading recommendation: Read the transition-forest construction to distinguish global hitting time from local stationarity. Keep the lower-bound qualification and unresolved proof step visible.